

- The locale should be *en_US.UTF-8*
- Check it using the following commands:

```
$ cat /etc/default/locale
LANG=en_US.UTF-8
LANGUAGE="en_IN:en"
if not, then install the packages,
$ sudo apt-get install language-pack-en
$ sudo update-locale LANG=en_US.UTF-8
```

- Also check the version of Ubuntu:

```
$ cat /etc/lsb_release
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=14.04
DISTRIB_CODENAME=trusty
DISTRIB_DESCRIPTION="Ubuntu 14.04.2 LTS"
```

Once the checks are over, here is how to go about the installation of BigBlueButton.

Server update

Check if *trusty multiverse* is there in the *sources.list*; if not, add the following:

```
$ grep "multiverse" /etc/apt/sources.list
```

The output should have the following:

```
deb http://in.archive.ubuntu.com/ubuntu/ trusty multiverse
```

If the above line is not there, you can add the following command:

```
$ echo "deb http://us.archive.ubuntu.com/ubuntu/ trusty
multiverse" | sudo tee -a /etc/apt/sources.list
```

Now update the server before proceeding to the next step, with the following command:

```
$ sudo apt-get update
$ sudo apt-get dist-upgrade
```

Once the upgrading is over, restart the OS and let's move on to installing LibreOffice. Ubuntu comes with LibreOffice 4.2.7, and BigBlueButton uses LibreOffice 4.3 for document conversion, so the PPA for LibreOffice 4.3 version has to be added:

```
$ sudo apt-get install software-properties-common
$ sudo add-apt-repository ppa:libreoffice/libreoffice-4-3
```

Installing the key of BigBlueButton

The local machine needs to access the BigBlueButton server, so the key has to be installed. Here is the command for installing the key:

Operating system	Ubuntu 14.01, 64-bit
RAM	Minimum of 4GB, but 8GB preferred
CPU	A decent quad core CPU
Ports to be unlocked	80, 1925, 9123 (TCP ports), 16384-32768 (UDP)
Space requirements	500GB for recording and storage
Bandwidth	100Mbps
Server	Can be a bare metal version rather than a virtual machine

Table 1: Prerequisites for BigBlueButton installation

```
$ wget http://ubuntu.bigbluebutton.org/bigbluebutton.asc -O-
| sudo apt-key add -
$ echo "deb http://ubuntu.bigbluebutton.org/trusty-090/
bigbluebutton-trusty main" | sudo tee /etc/apt/sources.
list.d/bigbluebutton.list
$ sudo apt-get update
```

Installing ffmpeg

ffmpeg needs to be installed since it creates the playback components like audio, video, presentation, desktop sharing, etc. This tool needs other dependencies also, so a group of tools has to be installed. Some customisation is also needed in the OS. So the entire code is pasted in a shell script and the script runs the commands, one by one.

Simply copy-paste the following code into a *ffmpeg_install.sh* file:

```
sudo apt-get install build-essential git-core checkinstall
yasm texi2html libvorbis-dev libx11-dev libvp8-dev libxfixes-
dev zlib1g-dev pkg-config netcat libncurses5-dev

FFMPEG_VERSION=2.3.3

cd /usr/local/src
if [ ! -d "/usr/local/src/ffmpeg-${FFMPEG_VERSION}" ]; then
  sudo wget "http://ffmpeg.org/releases/ffmpeg-${FFMPEG_
VERSION}.tar.bz2"
  sudo tar -xjf "ffmpeg-${FFMPEG_VERSION}.tar.bz2"
fi

cd "ffmpeg-${FFMPEG_VERSION}"
sudo ./configure --enable-version3 --enable-postproc --enable-
libvorbis --enable-libvp8
sudo make
sudo checkinstall --pkgname=ffmpeg --pkgversion="5:${FFMPEG_
VERSION}" --backup=no --deldoc=yes --default
```

Now, run:

```
$ sh ffmpeg_install.sh
```